NCP SYSTEM DESIGN SERIES

NCP 1.0 System Design

Low-overhead control, exact lifecycle ownership, bounded resources, and ecosystem boundaries

NCP project design note 17 August 2026

PROPOSED B01 DESIGN • UNRELEASED 1.0.0-rc.1 • NO RELEASE AUTHORITY

Abstract

This report presents the proposed high-level architecture for the unreleased NCP `1.0.0-rc.1` candidate. It separates control, perception, action, and observation traffic. It also defines exact identity, lifecycle, retry, freshness, and resource boundaries. The selected runtime prepares immutable context once. A hot frame is bounded, authenticated, decoded once, located exactly, admitted by one short owner transition, and handed to a finite plane queue.

The mathematical sections derive each symbolic bound from named variables and assumptions. They cover latency decomposition, finite memory, single-flight lifecycle ownership, exact retry, checked ordinals, queue isolation, fixed-layout work, step windows, and command freshness. These equations are design accounting. They are not measured performance or operational qualification.

The report is informative B01 material. It does not accept an ADR, allocate B03 values, change the normative wire, certify security or physical safety, qualify an ecosystem peer, or authorize publication. NCP remains release-blocked until every declared gate has exact evidence.

Contents

1 Reading contract and release boundary

Repository HEAD is the unreleased and release-blocked `1.0.0-rc.1` candidate. Its wire is `1.0`. Its compact protobuf contract hash is `163acc57d8a62b66`. The latest immutable release is `v0.8.0`, which uses a different wire.

This report describes the selected target for B01 review. The complete target is not the current runtime. B02 must authorize a deliberate rebaseline. B03 must allocate exact identities and numeric limits. Dependency-ready implementation tasks must then change each normative and generated surface together.

CLAIM BOUNDARY

Local source review, equations, diagrams, PDF reproduction, tests, and semantic models cannot establish live security, independent interoperability, or physical safety. They cannot establish performance qualification, signed provenance, consumer qualification, or release readiness.

The design uses three evidence states:

1. **Selected target** means proposed architecture for B01 review.
2. **Current prototype** means source that explores only part of that target.
3. **Implemented contract** requires authorized rebaseline, allocation, generation, implementation, and exact conformance evidence.

This report presents the selected target at the high-level design boundary. It does not complete the runtime or protocol. A diagram or equation cannot promote the target to an implemented contract.

2 System at a glance

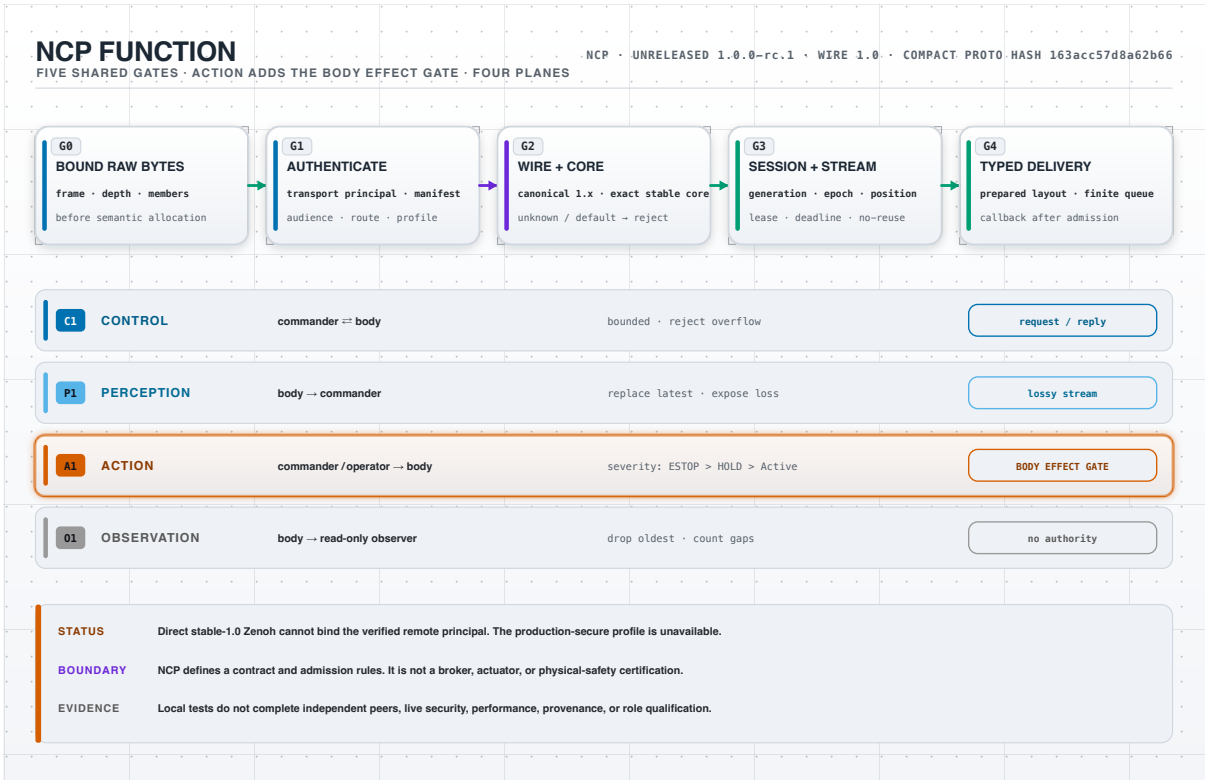

Figure 1: NCP admission and typed-delivery overview.

Figure 1 has one left-to-right reading order. A receiver first bounds raw input. It then authenticates ingress and checks the exact candidate wire identity. Session and stream state are checked before typed delivery. These are five shared gates. Action traffic evaluates a sixth, conditional body-owned effect predicate before software admission can succeed.

The four planes have different workload and overload properties:

Table 1: Plane ownership and overload policy.

Plane	Primary publisher	Ownership target	Overload result
Control	Commander or body	Request owner with exact idempotency context	Reject before effect
Perception	Body	Source stream owner	Replace latest and expose loss
Action	Commander or enrolled emergency source	Body owner and executor gate	Preserve restrictive priority
Observation	Body	Grant-bound projection owner	Drop oldest and count gaps

NCP defines contracts, admissions, libraries, and transport bindings. NCP is not a broker, actuator, or physical-safety certification.

3 Notation, units, and proof discipline

All counts are nonnegative integers. All byte quantities use bytes, not characters. All local time values use one receiver-owned monotonic clock incarnation. A digest is a content locator or binding. A digest does not replace an exact byte comparison when the design requires exact bytes.

Table 2: Core mathematical notation.

Symbol	Domain or unit	Meaning
C_x	nonnegative integer elements	Declared capacity of bounded store or queue x
B_x	nonnegative integer bytes per unit	Maximum portable charge for one capacity unit
B_0	nonnegative integer bytes	Fixed owner and store overhead
J_x	finite nonnegative local units	Profile-bound local allocator charge
T_i	nonnegative receiver duration	Duration of hot-path stage i
U_i	finite nonnegative receiver duration	Declared upper bound for stage i
o	finite nonnegative integer	Issuer operation ordinal
p	finite nonnegative integer	Stream or step position
d	nonnegative receiver monotonic tick	Exclusive deadline
$\|\|b\|\|$	nonnegative integer bytes	Exact length of byte string b

Each displayed inequality states its assumptions. Checked arithmetic rejects overflow before state mutation. A finite expression is not operational until B03 selects every input bound.

4 Trust topology and ordered admission

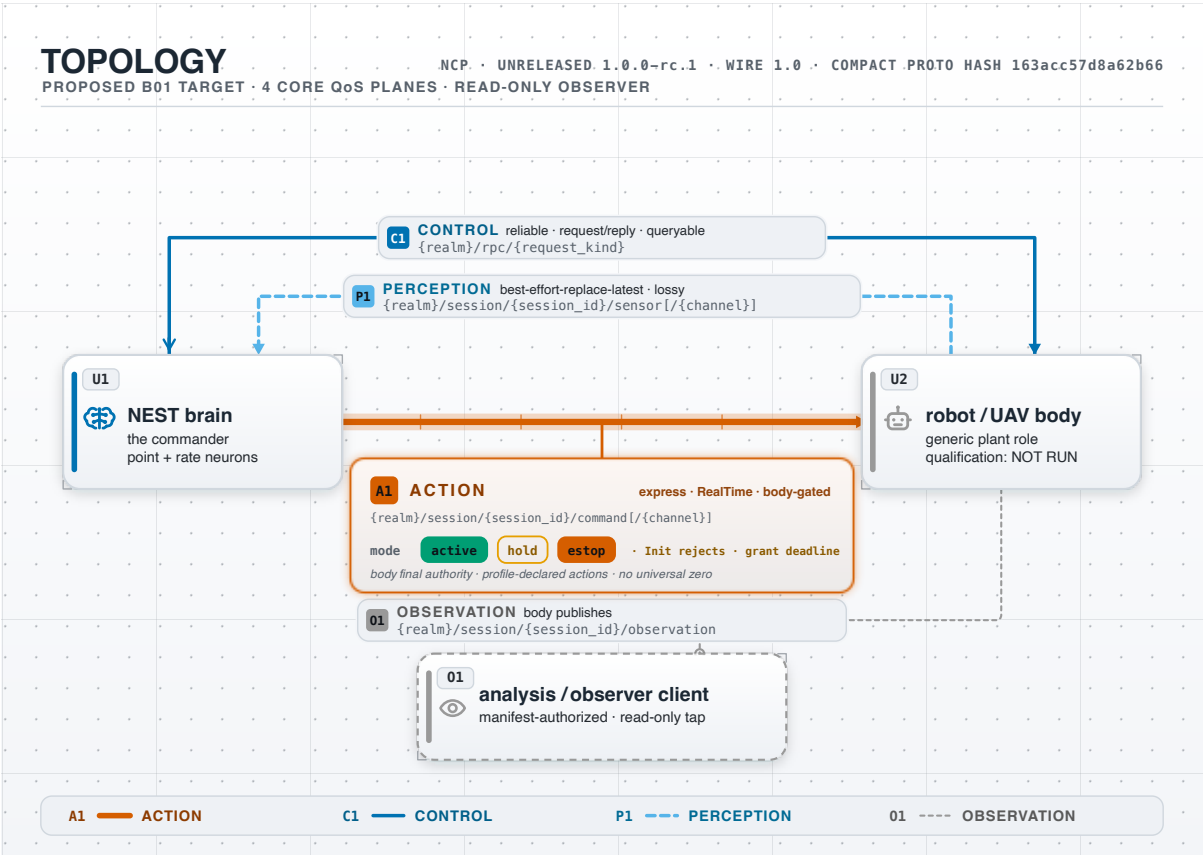

Figure 2: NCP commander-body topology and trust boundary.

The topology figure separates direct authenticated ingress from explicit forwarding. A direct peer must bind the verified transport principal. A forwarder terminates one trust context and creates another authenticated context. Copied identity bytes are not transport evidence.

Let f be one received frame. Let $q_j(f)$ be the Boolean result of admission gate j. The ordered gate set is

$$\mathcal{G} = (q_0, q_1, q_2, q_3, q_4, q_5) = (q_{\text{raw}}, q_{\text{principal}}, q_{\text{wire}}, q_{\text{session}}, q_{\text{typed}}, q_{\text{effect}}). \tag{1}$$

For a non-action frame, $q_{\text{effect}}(f) = \text{true}$ by definition. For an action frame, that predicate means body-owned software admission at the effect boundary. The complete admission predicate is the conjunction

$$A(f) = \bigwedge_{j=0}^{5} q_j(f). \tag{2}$$

The implementation does not evaluate equation (2) as an unordered set. It evaluates the prefix predicates

$$A_{-1}(f) = \text{true}, \tag{3}$$

$$A_j(f) = A_{j-1}(f) \wedge q_j(f), \qquad 0 \le j \le 5. \tag{4}$$

Gate $j+1$ can run only when $A_j(f)$ is true. Therefore raw bounds precede identity-sensitive diagnostics. The first false gate supplies the closed rejection class.

Derivation 4.1 (First-failure order)**.** If at least one gate is false, let $k = \min\{j \mid q_j(f) = \text{false}\}$. Equation (4) gives $A_{k-1}(f) = \text{true}$ and $A_k(f) = \text{false}$. No q_j for $j > k$ is evaluated. The receiver therefore emits no diagnostic that depends on an unauthenticated or unbounded later field.

The effect gate applies only to action. Other planes terminate at typed delivery and their finite queue. The body remains the final software authority before an actuator boundary.

5 Prepared runtime and hot-path latency

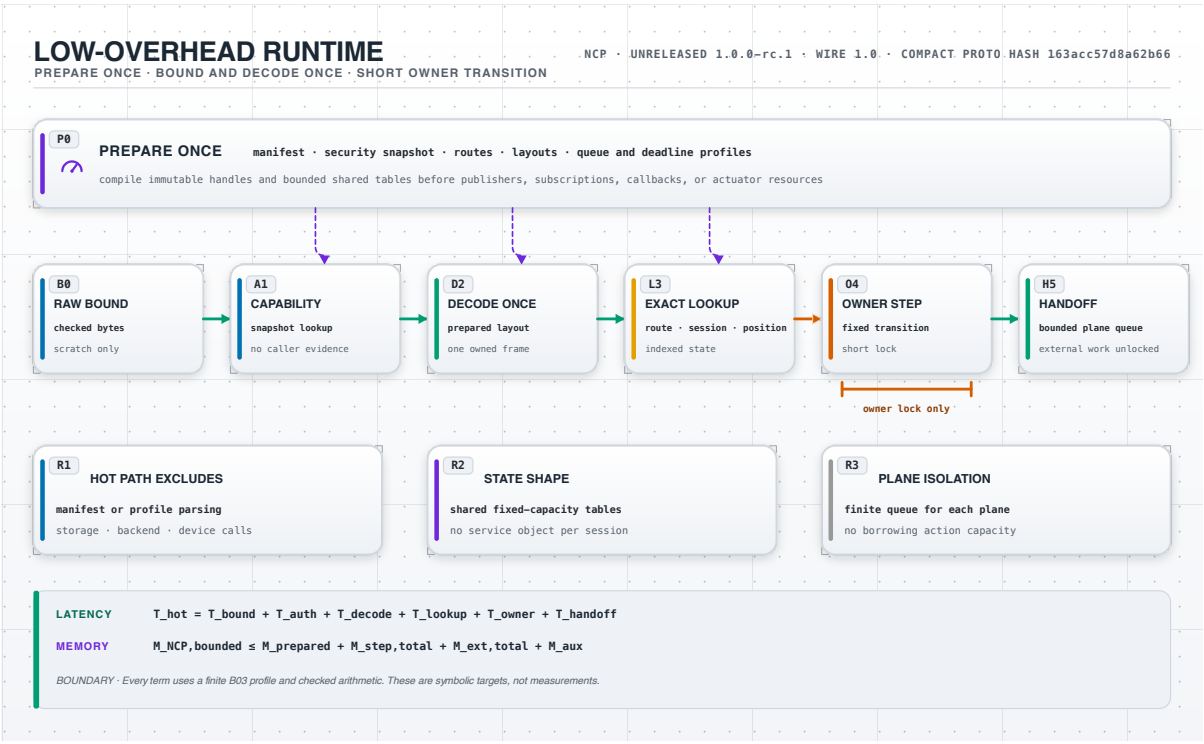

Figure 3: Prepared NCP runtime and symbolic cost model.

Preparation validates deployment input and compiles immutable handles. It reserves shared state before publishers, subscriptions, callbacks, or actuator resources are exposed. The repeated frame path then contains six local stages.

5.1 Stage intervals

Let the ordered stages be

$$\mathcal{I} = (\text{bound}, \text{auth}, \text{decode}, \text{lookup}, \text{owner}, \text{handoff}). \tag{5}$$

For each stage $i \in \mathcal{I}$, let t_i^{in} and t_i^{out} be receiver monotonic ticks from one clock incarnation. A cross-incarnation pair is invalid. Define the local duration

$$T_i = t_i^{\text{out}} - t_i^{\text{in}} \geq 0. \tag{6}$$

The serial accounting model assumes

$$t_i^{\text{out}} = t_{i+1}^{\text{in}} \tag{7}$$

for adjacent stages. The equality is an accounting boundary. It does not require a thread or process boundary.

Starting with the first input tick and ending at handoff gives

$$T_{\text{hot}} = t_{\text{handoff}}^{\text{out}} - t_{\text{bound}}^{\text{in}} \tag{8}$$

$$= \sum_{i \in \mathcal{I}} \left(t_i^{\text{out}} - t_i^{\text{in}} \right) \tag{9}$$

$$= T_{\text{bound}} + T_{\text{auth}} + T_{\text{decode}} + T_{\text{lookup}} + T_{\text{owner}} + T_{\text{handoff}}. \tag{10}$$

Equation (9) follows by inserting the adjacent boundary ticks from equation (7). Every internal tick occurs once with each sign, so the internal terms cancel.

5.2 Worst-case design bound

Suppose the selected profile supplies a finite stage ceiling U_i such that

$$0 \le T_i \le U_i \qquad \text{for each } i \in \mathcal{I}. \tag{11}$$

Monotonicity of addition gives

$$T_{\text{hot}} = \sum_{i \in \mathcal{I}} T_i \le \sum_{i \in \mathcal{I}} U_i = U_{\text{hot}}. \tag{12}$$

This is a conditional design bound. B03 must select each U_i. Later measurement must show that the implementation satisfies those ceilings on the qualified platform.

Transport delay precedes $t^{\text{in}}_{\text{bound}}$. Queued work, application work, and device work follow $t^{\text{out}}_{\text{handoff}}$. Those durations are outside T_{hot}.

5.3 Allocation and copy accounting

Let $a_i, c_i \in \mathbb{Z}_{\ge 0}$. The value a_i is the number of heap allocations caused by stage i after preparation. The value c_i is the number of full-payload copies. The totals are

$$A_{\text{hot}} = \sum_{i \in \mathcal{I}} a_i, \tag{13}$$

$$C_{\text{hot}} = \sum_{i \in \mathcal{I}} c_i. \tag{14}$$

For a fixed-layout compact frame with a caller-owned contiguous receive buffer, the target is

$$A_{\text{hot}} = 0, \qquad C_{\text{hot}} = 0 \tag{15}$$

between ingress and typed handoff. A transport can still own one bounded receive buffer. A segmented transport can make one bounded flattening copy. If preparation reserved that buffer, then $A_{\text{hot}} = 0$ and $C_{\text{hot}} \le 1$. An on-demand flattening buffer also adds one allocation and is outside the zero-allocation target.

Equation (15) is a target, not current evidence. Compatibility JSON can allocate within declared raw and semantic budgets.

6 Finite memory and queue isolation

6.1 Store-by-store derivation

Let $\mathcal{S} = \{s, c, r, w\}$ identify active session slots, prepared contexts, retained terminal results, and optional waiters. Store x has a profile capacity C_x. One capacity unit has a maximum charged size B_x bytes.

For every $x \in \mathcal{S}$, B03 selects $C_x, B_x \in \mathbb{Z}_{\ge 0}$. It also selects $B_0 \in \mathbb{Z}_{\ge 0}$. These are declared reservation charges, not sampled heap usage.

The portable charge B_x includes each specified variable component that scales with capacity. This includes content, backing storage, index entries, and metadata. It excludes runtime-specific allocator overhead, which belongs to the separate J_x domain. Let B_0 cover fixed portable owner and store overhead that does not scale with a declared capacity.

If store x reserves space for C_x units, then

$$M_x \le C_x B_x. \tag{16}$$

The product has units

$$[C_x B_x] = [\text{elements}]\,[\text{bytes/element}] = [\text{bytes}]. \tag{17}$$

The stores are disjoint accounting domains. Adding equation (16) across the four domains gives

$$\begin{aligned} M_{\text{lifecycle}} &\le B_0 + \sum_{x \in \mathcal{S}} M_x && (18)\\ &\le B_0 + \sum_{x \in \mathcal{S}} C_x B_x && (19)\\ &= B_0 + C_s B_s + C_c B_c + C_r B_r + C_w B_w. && (20) \end{aligned}$$

If waiters are disabled, $C_w = 0$. The bound still includes the fixed overhead B_0. This avoids the false conclusion that an empty preallocated table has zero memory.

All products and sums use checked arithmetic. Choose one fixed ordering $(x_1, \ldots, x_{|\mathcal{S}|})$ of $\mathcal{S}$. For $1 \le k \le |\mathcal{S}|$, define the partial sum

$$P_k = B_0 + \sum_{j=1}^{k} C_{x_j} B_{x_j}. \tag{21}$$

The receiver computes each product first. It then computes P_k in a fixed order. Any multiplication or addition overflow rejects the profile before allocation.

6.2 Portable and local allocation domains

Portable bytes do not prove a physical heap bound for every language runtime. Let $J_x(C_x)$ be the local allocator charge selected for store x. All J_x values and J_0 use one runtime-profile accounting unit. A local implementation independently enforces

$$J_{\text{local}} \le J_0 + \sum_{x \in \mathcal{S}} J_x(C_x). \tag{22}$$

The receiver admits a profile only when both the portable byte bound and the local charge bound pass. The function J_x must be finite and monotone over the permitted capacity range. It can include container nodes, capacity slack, handles, and allocator metadata. A JavaScript implementation must describe J_{local} as an abstract charge unless it controls the complete typed-array representation.

6.3 Plane queue envelope

Let $\mathcal{D}_{\text{plane}} = \{\text{ctl}, \text{per}, \text{act}, \text{obs}, \text{ext}\}$ identify control, perception, action, observation, and extension queues. Queue p has count capacity Q_p and aggregate byte capacity D_p. Let n_p be its retained-item count.

For every $p \in \mathcal{D}_{\text{plane}}$, B03 selects $Q_p \in \mathbb{Z}_{>0}$ and $D_p, B_{p,\text{meta}}, G_p \in \mathbb{Z}_{\ge 0}$. The value G_p is the largest representable visible-loss count. The current loss count satisfies $0 \le g_p \le G_p$. The live item count n_p and every byte-string length are nonnegative integers. The metadata charge covers the complete portable per-item metadata. Define the retained byte charge

$$S_p = \sum_{k=1}^{n_p} \|b_{p,k}\|. \tag{23}$$

The current queue invariant is $0 \le n_p \le Q_p$ and $0 \le S_p \le D_p$. The plane policy examines a new item b without retaining it. It either rejects or selects one deterministic allowable victim set $E_p(b) \subseteq \{1, \ldots, n_p\}$. Define

$$e_p(b) = |E_p(b)|, \qquad R_p(b) = \sum_{k \in E_p(b)} \|b_{p,k}\|. \tag{24}$$

Control and extension select the empty set under reject-on-overflow policy. Perception can select the current replaceable latest item. Observation selects the oldest items in order. Action can select only work that its restrictive severity policy permits it to displace. If no allowable victim set makes room, the incoming item rejects and the queue does not change.

Before mutation, the owner computes the survivor state

$$n_p^- = n_p - e_p(b), \qquad S_p^- = S_p - R_p(b). \tag{25}$$

The subset relation gives $0 \le e_p(b) \le n_p$ and $0 \le R_p(b) \le S_p$. The subtraction and every later sum use checked arithmetic. Admission requires

$$0 \le n_p^- < Q_p, \tag{26}$$

$$0 \le S_p^- \quad \wedge \quad S_p^- + \|b\| \le D_p. \tag{27}$$

Before mutation, the owner also computes

$$g_p' = \text{checked\_add}(g_p, e_p(b)) \le G_p. \tag{28}$$

If the checked successor is unavailable, the owner seals or retires the affected stream and leaves the queue unchanged.

One atomic owner transition reserves the incoming item, records the visible loss, removes the selected victims, and installs the new item. The resulting state is

$$n_p' = n_p^- + 1 \le Q_p, \tag{29}$$

$$S_p' = S_p^- + \|b\| \le D_p. \tag{30}$$

A count limit alone is insufficient because item size varies. The queue reserves both one count unit and the complete byte charge before it copies or retains the incoming item.

Let $B_{\text{queue},0} \in \mathbb{Z}_{\ge 0}$ be the fixed portable charge for queue owners and indexes that does not scale with a declared queue capacity. The total portable queue envelope is

$$M_{\text{queues}} \le B_{\text{queue},0} + \sum_{p \in \mathcal{D}_{\text{plane}}} \left(D_p + Q_p B_{p,\text{meta}}\right). \tag{31}$$

The lifecycle stores and plane queues are disjoint portable accounting domains. Combining equations (19) and (31) gives

$$M_{\text{prepared}} \le B_0 + \sum_{x \in \mathcal{S}} C_x B_x + B_{\text{queue},0} + \sum_{p \in \mathcal{D}_{\text{plane}}} \left(D_p + Q_p B_{p,\text{meta}}\right). \tag{32}$$

Let $J_{\text{queue},0}$ be the fixed local-runtime charge for the queue owners. Let $J_p^{\text{queue}}(Q_p, D_p)$ be queue p's finite local-runtime charge. It is monotone in both arguments and uses the same runtime-profile accounting unit as J_0. The independently checked local queue bound is

$$J_{\text{queues}} \le J_{\text{queue},0} + \sum_{p \in \mathcal{D}_{\text{plane}}} J_p^{\text{queue}}(Q_p, D_p). \tag{33}$$

The lifecycle stores and plane queues are disjoint accounting domains. Their combined local charge therefore satisfies

$$J_{\text{prepared}} = J_{\text{local}} + J_{\text{queues}} \tag{34}$$

$$\le J_0 + \sum_{x \in \mathcal{S}} J_x(C_x) + J_{\text{queue},0} + \sum_{p \in \mathcal{D}_{\text{plane}}} J_p^{\text{queue}}(Q_p, D_p). \tag{35}$$

No queue can borrow another plane's D_p, Q_p, or local charge. Therefore observation pressure cannot consume reserved action capacity.

Limitation 6.1 (Meaning of the memory bounds)**.** Equations (20), (22), and (35) exclude transport buffers and unrelated process memory. They also exclude device memory. They become operational only after B03 selects exact numeric profiles.

7 Session lifecycle mutation owner

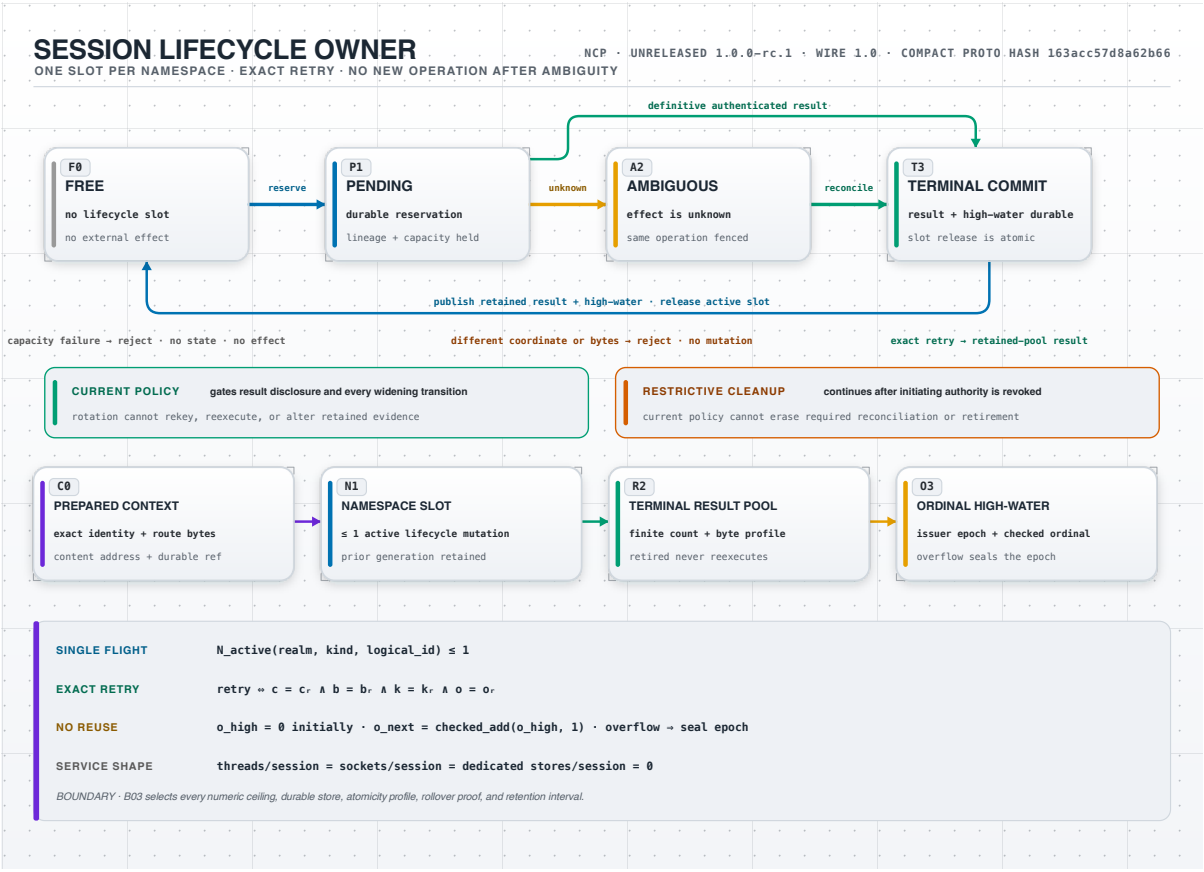

Figure 4: Single-flight session lifecycle owner and exact retry model.

One bounded slot serializes open, replacement, and close for one namespace. The namespace is the tuple

$$\nu = (\text{realm}, \text{session\_kind}, \text{logical\_id}). \tag{36}$$

The active states are `PENDING` and `AMBIGUOUS`. A terminal record lives in a separate retained-result pool.

7.1 Single-flight invariant

Let $\mathcal{O}_\nu$ be all lifecycle records for namespace ν. Define

$$N_{\text{active}}(\nu) = \sum_{u \in \mathcal{O}_\nu} \mathbf{1}[\text{state}(u) \in \{\text{PENDING}, \text{AMBIGUOUS}\}]\,. \tag{37}$$

The owner has one active slot for ν. Each transition into an active state first proves that the slot is free. Therefore

$$0 \le N_{\text{active}}(\nu) \le 1. \tag{38}$$

The upper bound is structural. A mutex alone is insufficient across processes. A shared writer must also bind one fenced writer epoch.

7.2 Durable transition order

Let L be the prior lineage. Let E be the external effect state. Let R be the reserved terminal-result cell. Let K be the prepared-context reference. The transition from free to pending is ordered as

$$(\text{FREE}, L) \xrightarrow{\text{reserve}(R,K,L)} (\text{PENDING}, L, R, K) \xrightarrow{\text{external work}} E. \tag{39}$$

The first edge is crash-consistent. External work cannot start before that edge is durable. Capacity failure therefore leaves no partial slot and starts no external effect.

An unknown result does not create a new operation:

$$(\text{PENDING}, c) \xrightarrow{\text{unknown effect}} (\text{AMBIGUOUS}, c). \tag{40}$$

The immutable coordinate c remains unchanged. Reconciliation queries the same backend coordinate and then moves the same record to terminal commit.

Terminal commit has the order

$$\text{publish}(r, o_{\text{high}}) \prec \text{release}(\text{active slot}), \tag{41}$$

where $\prec$ means *commits before*. A later operation cannot observe a free slot without also observing the retained result and advanced high-water.

7.3 Exact retry relation

Let the retained operation be

$$u_r = (c_r, b_r, k_r, o_r). \tag{42}$$

Here c_r is the immutable coordinate. The value b_r is the exact request bytes. The value k_r is the exact admission-context bytes. The value o_r is the issuer ordinal.

For retry input $u = (c, b, k, o)$, define

$$R_{\text{exact}}(u, u_r) \iff (c = c_r) \wedge (b = b_r) \wedge (k = k_r) \wedge (o = o_r). \tag{43}$$

$$R_{\text{conflict}}(u, u_r) \iff \neg R_{\text{exact}}(u, u_r). \tag{44}$$

Equality of b and k is byte-for-byte equality. A digest can select a candidate record. The receiver still exact-compares the retained bytes before it returns or mutates state.

The relation in equation (43) is reflexive, symmetric, and transitive because each component uses exact equality. Therefore it partitions retry inputs into equivalence classes. Only the retained class can observe the retained result.

7.4 Checked ordinal and no-reuse proof

Let $O_{\max} \in \mathbb{Z}_{>0}$ be finite for one issuer epoch. Define the assignable ordinal domain and initial high-water sentinel by

$$\mathcal{D}_o = \{1, 2, \ldots, O_{\max}\}, \qquad o_{\text{high}} \in \{0\} \cup \mathcal{D}_o. \tag{45}$$

The value $o_{\text{high}} = 0$ means that the epoch has committed no operation ordinal. Otherwise, o_{high} is the largest committed ordinal. For a terminal high-water o_{high}, the successor exists only when

$$o_{\text{high}} < O_{\max}. \tag{46}$$

Under that precondition, the next ordinal is

$$o_{\text{next}} = \text{checked\_add}(o_{\text{high}}, 1). \tag{47}$$

Checked addition gives

$$o_{\text{next}} = o_{\text{high}} + 1 > o_{\text{high}}, \qquad o_{\text{next}} \in \mathcal{D}_o. \tag{48}$$

If equation (46) is false, the epoch seals. The ordinal never wraps or saturates. An input $o \leq o_{\text{high}}$ returns retained, retired, or unavailable. It never executes again.

7.5 No dedicated service per session

Shared fixed-capacity tables serve many namespaces. The design target is

$$N_{\text{thread/session}} = N_{\text{socket/session}} = N_{\text{dedicated store/session}} = 0. \tag{49}$$

Equation (49) does not remove session data. The data lives in bounded shared owners and stores. The lifecycle memory bound remains equation (20).

8 Streams, positions, and fixed-layout work

An accepted stream declaration compiles one immutable context. The context binds route, publisher incarnation, epoch, frame class, encoding, layout, position range, replay profile, and overload policy.

8.1 Position allocation

Let $P_{\max} \in \mathbb{Z}_{>0}$ be the selected maximum position for one stream epoch. Define the assignable domain and its initial high-water sentinel by

$$\mathcal{D}_p = \{1, 2, \ldots, P_{\max}\}, \qquad p_{\text{high}} \in \{0\} \cup \mathcal{D}_p. \tag{50}$$

The value $p_{\text{high}} = 0$ means that the epoch has assigned no position. Otherwise, p_{high} is the largest assigned position. A successor can exist only when

$$p_{\text{high}} < P_{\max}. \tag{51}$$

Under that precondition, assignment uses

$$p_{\text{candidate}} = \text{checked\_add}(p_{\text{high}}, 1). \tag{52}$$

Checked addition and equation (51) give

$$p_{\text{candidate}} = p_{\text{high}} + 1 > p_{\text{high}}, \qquad p_{\text{candidate}} \in \mathcal{D}_p. \tag{53}$$

If equation (51) is false, the owner seals the stream epoch. It does not wrap, saturate, or allocate a position in a new epoch implicitly. The allocator commits the candidate before later local encode, queue, or transport work. A later failure creates a visible gap. The allocator never reuses the candidate.

Let $\mathcal{P}_{\text{assigned}}$ be the set of assigned positions and $\mathcal{P}_{\text{published}}$ the set of published positions. Then

$$\mathcal{P}_{\text{published}} \subseteq \mathcal{P}_{\text{assigned}}. \tag{54}$$

The visible gap set is

$$\mathcal{P}_{\text{gap}} = \mathcal{P}_{\text{assigned}} \setminus \mathcal{P}_{\text{published}}. \tag{55}$$

Because assignment is monotone, no element of $\mathcal{P}_{\text{gap}}$ can later identify a different payload.

8.2 Fixed-layout frame size

Let $n_v, N_v, n_g, N_g, H_f \in \mathbb{Z}_{\geq 0}$ and $w_v \in \mathbb{Z}_{>0}$. A prepared frame contains n_v numeric slots. Each encoded value uses w_v bytes. The fixed header uses H_f bytes. The compact-sensor group count is n_g, bounded by N_g. Its mandatory availability storage is

$$A_f(n_g) = \left\lceil \frac{n_g}{8} \right\rceil. \tag{56}$$

A sensor layout requires $n_g \geq 1$. Its nonempty ordered groups partition every sensor slot exactly once. A non-sensor fixed layout uses $n_g = 0$ and no availability bytes. The complete frame length is

$$L_{\text{frame}} = H_f + A_f(n_g) + n_v w_v. \tag{57}$$

Suppose the profile selects $0 \leq n_v \leq N_v$ and $0 \leq n_g \leq N_g$. Then

$$L_{\text{frame}} = H_f + A_f(n_g) + n_v w_v \tag{58}$$

$$\leq H_f + \left\lceil \frac{N_g}{8} \right\rceil + N_v w_v = L_{\text{frame}}^{\max}. \tag{59}$$

The receiver checks n_v and the complete length before it decodes values. The names, units, ranges, and offsets live in the prepared layout. The product and sum use checked arithmetic. An unavailable product or sum rejects before value decoding. Names and units are not repeated or allocated per frame.

Stable 1.0 fixes one bit per ordered group, least-significant-bit-first order, 1 = AVAILABLE, 0 = UNAVAILABLE, and zero high padding. A compact sensor layout fixes group membership and profile-specific unavailable placeholder bits. Every group is decided once. Available groups contain every finite in-range value. Unavailable groups forbid caller values. The decoder returns typed unavailability and never exposes placeholders as observations. The complete frame digest covers the header, bitmap, and every scalar byte.

The header, bitmap, and scalar storage form one queue item. They share one position, digest, supersession decision, replay record, drop result, and receipt. Received available zero, received unavailable, malformed frame, transport gap, and whole-frame loss remain distinct. Compatibility JSON carries the same semantic availability and never infers it from zero or omission.

For a direct linear decode, the work count satisfies

$$W_{\text{codec}}(n_v) = \alpha + \beta n_v, \tag{60}$$

where $\alpha, \beta \in \mathbb{Z}_{\geq 0}$ are fixed operation counts for header work and work per value. This equation is a structural operation count. It is not a time measurement.

9 Simulation-step execution

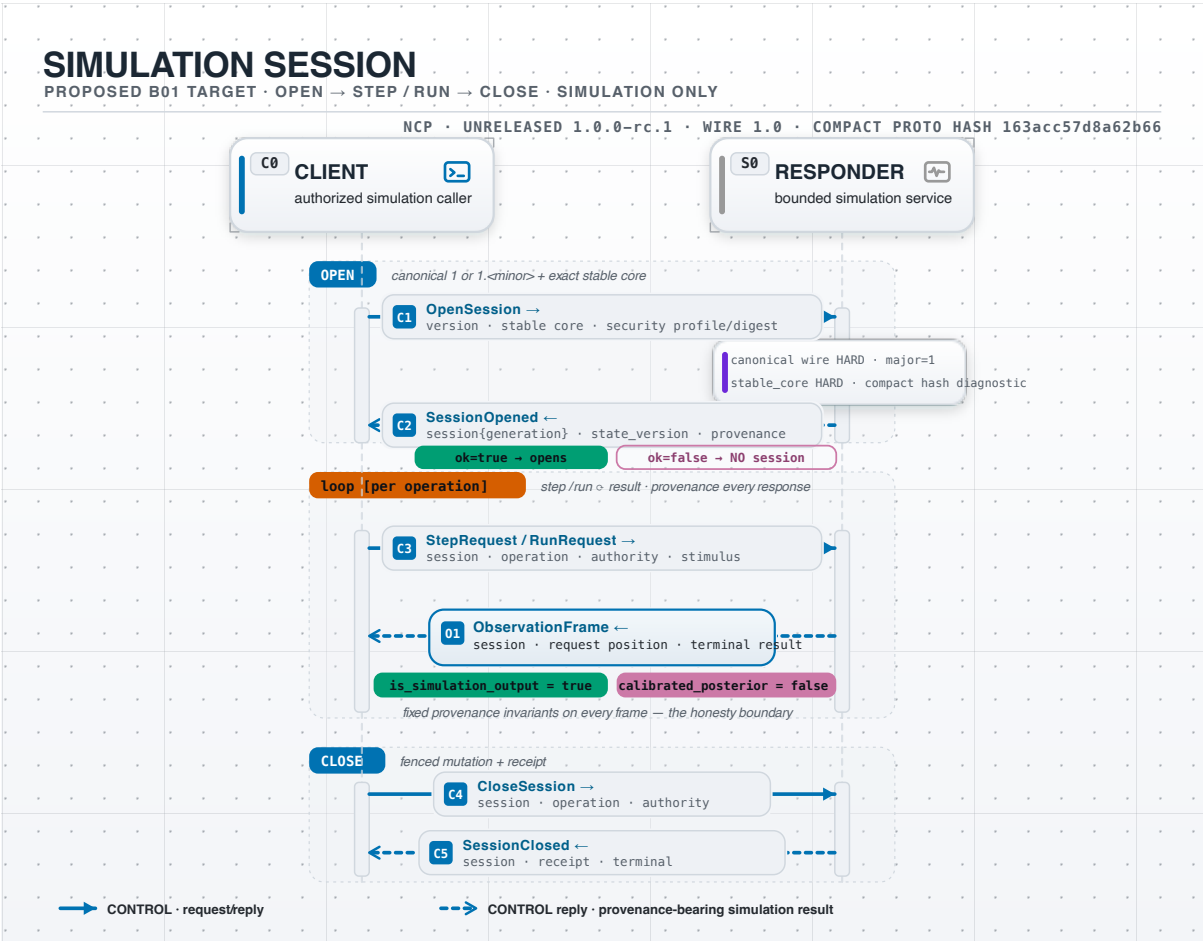

Figure 5: Simulation-step request, execution, and retained-response sequence.

Lifecycle control stays on bounded canonical JSON. Repeated simulation steps use one prepared fixed-layout request and response pair. The pair binds a live simulation generation, authority grant, exact request position, advance interval, input layout, output layout, and retained result.

9.1 Window and memory derivation

Let $C_{\text{step}} \in \mathbb{Z}_{>0}$ be the deployment maximum for prepared pipeline capacity. Let $n_{\text{open}} \in \mathbb{Z}_{\geq 0}$ be the number of reserved requests that are not terminal. Admission requires

$$0 \leq n_{\text{open}} < C_{\text{step}}. \tag{61}$$

The strict upper relation leaves one free reservation for the new request. After reservation, the invariant becomes

$$0 \leq n_{\text{open}} + 1 \leq C_{\text{step}}. \tag{62}$$

Let B_{req} and B_{res} be the maximum retained request and response bytes. Let $B_{\text{step,meta}}$ be the maximum per-slot metadata bytes. Let $B_{\text{step},0}$ be fixed window overhead. All four byte charges are nonnegative integers. These charges are deployment maxima. Every accepted step-window profile is at or below them. Pre-reserving both directions gives

$$M_{\text{step}} \leq B_{\text{step},0} + C_{\text{step}} \left(B_{\text{req}} + B_{\text{res}} + B_{\text{step,meta}} \right). \tag{63}$$

Let $J_{\text{step},0}$ and $J_{\text{step,slot}}$ be finite nonnegative charges in one selected local-runtime unit. The slot charge includes its request, response, index, and container representation. These charges are deployment maxima for that runtime profile. The separate local bound is

$$J_{\text{step}} \le J_{\text{step},0} + C_{\text{step}} J_{\text{step,slot}}. \tag{64}$$

Each active simulation session owns at most one prepared step window. Each simultaneous window is charged to one distinct active session slot. Let n_{win} be the number of simultaneous windows. The lifecycle capacity therefore gives

$$0 \le n_{\text{win}} \le C_s. \tag{65}$$

Let $M_{\text{step}}^{(a)}$ and $J_{\text{step}}^{(a)}$ be the portable and local charges for window a. Summing the per-window bounds and then applying the count bound gives

$$\begin{aligned} M_{\text{step,total}} &= \sum_{a=1}^{n_{\text{win}}} M_{\text{step}}^{(a)} \\ &\le n_{\text{win}} \left[B_{\text{step},0} + C_{\text{step}} \left(B_{\text{req}} + B_{\text{res}} + B_{\text{step,meta}} \right) \right] \\ &\le C_s \left[B_{\text{step},0} + C_{\text{step}} \left(B_{\text{req}} + B_{\text{res}} + B_{\text{step,meta}} \right) \right], \end{aligned} \tag{66}$$

$$\begin{aligned} J_{\text{step,total}} &= \sum_{a=1}^{n_{\text{win}}} J_{\text{step}}^{(a)} \\ &\le n_{\text{win}} \left[J_{\text{step},0} + C_{\text{step}} J_{\text{step,slot}} \right] \\ &\le C_s \left[J_{\text{step},0} + C_{\text{step}} J_{\text{step,slot}} \right]. \end{aligned} \tag{67}$$

Every product and sum uses checked arithmetic. The profile checks the per-window and deployment-wide bounds before it exposes the compact step path. The owner keeps the session slot charged while its step window is active or retains a response. A close can instead move a longer-lived terminal response into the separately charged retained- result pool. That transfer and window release form one atomic owner transition. The response cell is reserved before execution. Backend work therefore cannot complete into an unbounded or unavailable result path.

9.2 Strict execution order

Let $e, p \in \mathbb{Z}_{\ge 0}$ be positions in one finite generation domain. The value e is the next executable position. A request at position p can start only when

$$p = e. \tag{68}$$

On definitive completion, the owner computes

$$e' = \text{checked\_add}(e, 1). \tag{69}$$

If the successor cannot be represented, the owner seals the generation and admits no later position. An unresolved backend call retires the generation. The owner does not infer order from arrival or from a FIFO waiter list.

9.3 Exact response correlation

Let $h_p = \text{digest}(b_p)$ index the exact request bytes at position p. The request store retains b_p until the terminal response is durable. The retained response is the tuple

$$r_p = (\text{generation}, p, h_p, \text{result}). \tag{70}$$

A response can satisfy only a waiter with the same generation, position, request digest, and exact request bytes. The digest locates the candidate. The receiver exact-compares b_p before it releases the response. A reply cannot consume the next FIFO waiter by arrival order.

10 Plant command admission and freshness

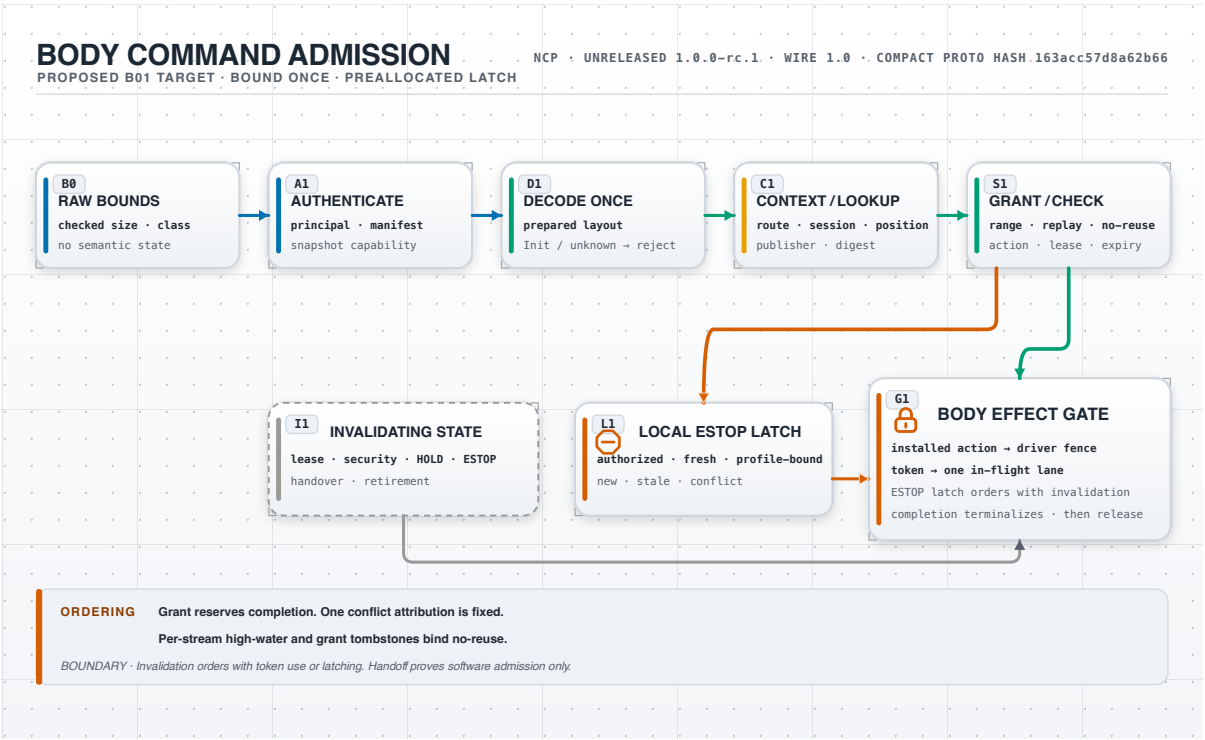

Figure 6: Body-owned plant command admission order.

The body issues live authority and command freshness. Serialized lease possession does not grant authority. Receiver arrival never starts or refreshes a remote command lifetime.

10.1 Receiver-owned deadline

Let $t_{\text{issue}} \in \mathbb{Z}_{\geq 0}$ be the receiver monotonic tick sampled when the body creates a freshness grant. Let $\Delta_{\max} \in \mathbb{Z}_{>0}$ be the finite profile lifetime in ticks. Both values use one receiver clock incarnation. The exclusive deadline is

$$d = \text{checked\_add}(t_{\text{issue}}, \Delta_{\max}). \tag{71}$$

If checked addition overflows, the body rejects the grant before it becomes visible. The grant is live at nonnegative receiver tick t from that same clock incarnation only when

$$t_{\text{issue}} \leq t < d. \tag{72}$$

An arrival tick t_{arrival} is audit evidence. Replacing t_{issue} with t_{arrival} would create a new lifetime after network delay. That replacement is forbidden.

Let a grant cover the contiguous integer position range $[p_0, p_1]$, where $0 \leq p_0 \leq p_1$. Position admission also requires

$$p_0 \leq p \leq p_1. \tag{73}$$

The position selects the grant. The frame does not repeat the grant, lease, or lifetime.

10.2 Active command predicate

Let the Boolean predicates be:

$$\begin{aligned} q_c &= \text{security snapshot and session are current},\\ q_l &= \text{the exact authority lease is live},\\ q_g &= \text{the freshness grant satisfies equations (72) and (73)},\\ q_s &= \text{stream epoch and position pass no-reuse checks},\\ q_p &= \text{plant profile accepts channel, arity, units, and range},\\ q_e &= \text{the bounded executor slot accepts the exact command}. \end{aligned}$$

An Active command is software-admitted only when

$$A_{\text{active}} = q_c \wedge q_l \wedge q_g \wedge q_s \wedge q_p \wedge q_e. \tag{74}$$

The equation stops at software admission. It does not state that a physical effect occurred.

10.3 Restrictive modes

HOLD requires an explicit known mode and a live holder. HOLD follows ordinary stream monotonicity. ESTOP can use the separately enrolled emergency lane when policy permits it. Neither mode carries a remote value vector or source coordinate. Their actions come from the installed plant profile.

Remote rejection and a body-local restrictive response are different results. A malformed or unauthorized remote request cannot claim that the remote HOLD or ESTOP effect succeeded.

11 Body effect boundary

The body owner creates one bounded executor-admission token. The token binds the exact command, plant profile, generation, authority state, freshness state, and executor domain incarnation. The actuator interface does not accept a raw map, transport message, or serialized lease.

Let a be the software-admission event and x a later physical effect. NCP can retain evidence for a. The protocol proof boundary is

$$\text{NCP} \nvdash (a \Longrightarrow x)\,. \tag{75}$$

The turnstile states that NCP does not prove the implication. It is not a claim that the effect is impossible. Device, mechanical, electrical, and plant-local safety systems remain outside the protocol proof boundary. A deployment needs its own safety case and independent interlock.

12 Extensions and bounded semantic envelopes

One extension message contains one complete bounded canonical-JSON envelope. Let $L, L_{\max} \in \mathbb{Z}_{>0}$ be the delivered envelope bytes and their installed maximum. Let $A, A_{\max} \in \mathbb{Z}_{\geq 0}$ be the attachment count and its installed maximum. Admission requires

$$0 < L \leq L_{\max}, \qquad 0 \leq A \leq A_{\max}. \tag{76}$$

For declared attachment lengths b_i and aggregate maximum B_{att}, admission also requires

$$\sum_{i=0}^{A-1} b_i \leq B_{\text{att}}. \tag{77}$$

The installed manifest selects one closed schema and canonicalization profile. It rejects unknown members, duplicate decoded keys, non-canonical numbers, unsupported encodings, and inline attachment bytes before callback work. Each reference binds an attachment ID, enrolled store ID, canonical object key, digest, byte length, media type, schema ID, and purpose. The selected ingress profile also binds the realm, activation, audience, manifest, and replay coordinate. A-direct uses the receiver-owned context. B-over-A uses the protected JWS envelope.

The installed manifest enrolls the exact HTTPS origin, resolved address set, TLS identity, scoped credential source, timeouts, media types, and byte limits. The wire cannot supply a URL, user information, query, fragment, local path, or redirect target. Fetches use no ambient credential. Redirects, DNS or address drift, private-address substitution, local-file resolution, and symlink traversal reject.

The receiver reserves all resources before it mints a one-use local fetch capability. That capability binds the envelope digest and exact reference. The receiver verifies the declared length and digest before semantic use. Missing, partial, oversized, mismatched, or unavailable bytes make the dependent branch unusable.

After every attachment verifies, the activation owner performs one atomic transition. It rechecks activation, security, principal, audience, manifest, realm, session, freshness, revocation, expiry, and resource currentness. The same transition consumes the one callback right. A cut during fetch prevents callback entry. NCP 1.0 defines no generic chunk-reassembly protocol.

Let $C_{\mathrm{ext}} \in \mathbb{Z}_{\geq 0}$ be the deployment maximum for all active and retained semantic-envelope slots in one activation. Let $B_{\mathrm{ext},0} \in \mathbb{Z}_{\geq 0}$ be the deployment maximum for its fixed portable charge. Let $H_{\mathrm{ext}}^{\max} \in \mathbb{Z}_{\geq 0}$ be the deployment maximum of the greater active or terminal portable metadata charge for one slot. Every accepted activation profile is at or below these maxima. The outer portable reservation satisfies

$$M_{\mathrm{ext,outer}} \leq B_{\mathrm{ext},0} + C_{\mathrm{ext}} \left(L_{\max} + H_{\mathrm{ext}}^{\max}\right). \tag{78}$$

Each installed extension activation consumes one distinct prepared-context unit. Let n_{ext} be the number of simultaneous extension activations. Equation (20) gives

$$0 \leq n_{\mathrm{ext}} \leq C_c. \tag{79}$$

Let $M_{\mathrm{ext,outer}}^{(a)}$ be activation a's portable envelope charge. Summing the per-activation bound gives

$$\begin{aligned} M_{\mathrm{ext,total}} &= \sum_{a=1}^{n_{\mathrm{ext}}} M_{\mathrm{ext,outer}}^{(a)} \\ &\leq n_{\mathrm{ext}} \left[B_{\mathrm{ext},0} + C_{\mathrm{ext}} \left(L_{\max} + H_{\mathrm{ext}}^{\max}\right)\right] \\ &\leq C_c \left[B_{\mathrm{ext},0} + C_{\mathrm{ext}} \left(L_{\max} + H_{\mathrm{ext}}^{\max}\right)\right]. \end{aligned} \tag{80}$$

Let $J_{\mathrm{ext},0}$ be the fixed per-activation local charge. Let $J_{\mathrm{ext,slot}}(L_{\max}, A_{\max})$ be a finite nonnegative slot charge in one selected local-runtime unit. The slot function is monotone in both arguments. The independent local bound is

$$J_{\mathrm{ext,outer}} \leq J_{\mathrm{ext},0} + C_{\mathrm{ext}} J_{\mathrm{ext,slot}}(L_{\max}, A_{\max}). \tag{81}$$

Let $J_{\mathrm{ext,outer}}^{(a)}$ be activation a's local envelope charge. The same count bound gives

$$\begin{aligned} J_{\mathrm{ext,total}} &= \sum_{a=1}^{n_{\mathrm{ext}}} J_{\mathrm{ext,outer}}^{(a)} \\ &\leq n_{\mathrm{ext}} \left[J_{\mathrm{ext},0} + C_{\mathrm{ext}} J_{\mathrm{ext,slot}}(L_{\max}, A_{\max})\right] \\ &\leq C_c \left[J_{\mathrm{ext},0} + C_{\mathrm{ext}} J_{\mathrm{ext,slot}}(L_{\max}, A_{\max})\right]. \end{aligned} \tag{82}$$

Every product, sum, and offset uses checked arithmetic. Overflow rejects the profile or activation before it allocates a prepared context or envelope slot. An activation context remains charged while any active envelope slot or terminal tombstone refers to it. The owner cannot release that context first. Therefore a retained tombstone cannot escape the C_c multiplier.

The receiver reserves envelope, parser, attachment, callback, and terminal-state budgets before semantic work. Exact replay is idempotent. Conflicting bytes terminalize the same slot. Canonical parsing starts only after authentication and complete reservation. The envelope bounds exclude transport scratch, attachment storage, parser state, callback state, and unrelated process memory. The auxiliary envelope accounts for those NCP-owned domains. B03 selects all finite inputs.

12.1 Complete bounded-state envelope

Let $\mathcal{D}_{\text{aux}} = \{\text{rx}, \text{json}, \text{attachment}, \text{extparse}, \text{callback}\}$ identify transport receive or flattening scratch, structured-JSON arenas, verified attachment storage, extension inner-parser arenas, and typed callback buffers. B03 selects one deployment-wide portable ceiling $B_y \in \mathbb{Z}_{\geq 0}$ and one finite monotone local charge $J_y(B_y)$ for every $y \in \mathcal{D}_{\text{aux}}$. It also selects fixed charges $B_{\text{aux},0}$ and $J_{\text{aux},0}$. The auxiliary envelopes are

$$M_{\text{aux}} \leq B_{\text{aux},0} + \sum_{y \in \mathcal{D}_{\text{aux}}} B_y, \tag{83}$$

$$J_{\text{aux}} \leq J_{\text{aux},0} + \sum_{y \in \mathcal{D}_{\text{aux}}} J_y(B_y). \tag{84}$$

These domains are disjoint from lifecycle, queue, step-window, and envelope storage. Adding the previously derived deployment bounds gives the complete NCP-owned bounded-state model

$$M_{\text{NCP,bounded}} \leq M_{\text{prepared}} + M_{\text{step,total}} + M_{\text{ext,total}} + M_{\text{aux}}, \tag{85}$$

$$J_{\text{NCP,bounded}} \leq J_{\text{prepared}} + J_{\text{step,total}} + J_{\text{ext,total}} + J_{\text{aux}}. \tag{86}$$

Every component is independently finite and checked before its owner becomes reachable. No domain can borrow another domain's reservation. These equations bound declared NCP-owned state, not code, thread stacks, device memory, unrelated process state, or a universally portable physical RSS. The sum applies no alias discount. A zero-copy view is charged to its backing owner. A copied buffer is a second charge. During a cross-owner transfer, both domains remain reserved until the destination commit preserves every required recovery byte and permits source release. The sum therefore covers the transfer high-water.

13 Ecosystem dependency direction

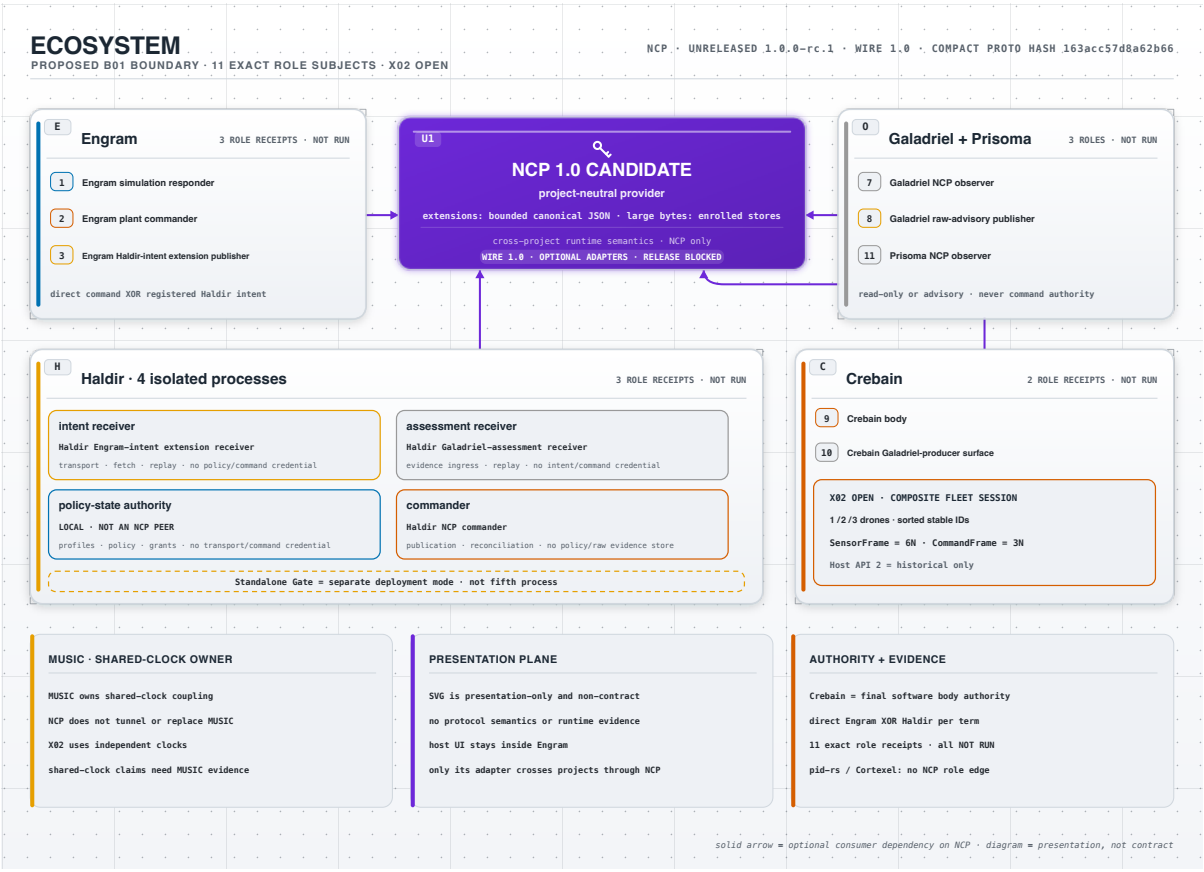

Figure 7: NCP provider and ecosystem dependency direction.

NCP owns the protocol, generated bindings, profiles, and package surfaces. Consumer projects pin NCP packages and implement thin role adapters. NCP does not import consumer application code.

Let $G = (V, E)$ be the repository dependency graph. An edge $(u, v) \in E$ means repository u depends on repository v. If $n \in V$ is NCP and $C \subset V$ is the set of consumers, the selected direction is

$$\forall c \in C : \ (c, n) \in E \quad \wedge \quad (n, c) \notin E. \tag{87}$$

The condition prevents a consumer-specific fork from becoming an NCP requirement.

13.1 Exact ecosystem role subjects

The ecosystem release matrix contains exactly eleven role subjects. Each subject needs a distinct installed-artifact and live-transport qualification receipt. All eleven qualifications remain **NOT RUN**.

Table 3: Exact ecosystem role subjects.

Role subject	Boundary
Engram simulation responder	Simulation session only
Engram plant commander	Direct plant commander
Engram Haldir-intent extension publisher	Registered extension publisher

Role subject	Boundary
Haldir NCP commander	Gated plant commander
Haldir Engram-intent extension receiver	Registered extension receiver
Haldir Galadriel-assessment receiver	Isolated assessment-extension receiver
Galadriel NCP observer	Read-only observer
Galadriel raw-advisory publisher	Isolated advisory-extension publisher
Crebain body	Plant responder and final software body authority
Crebain Galadriel-producer surface	Isolated sidecar-evidence publisher
Prisoma NCP observer	Read-only observer

13.2 Haldir process separation

Integrated Haldir uses four separate deployable processes. No integrated process activates two role surfaces. The policy-state authority is not an NCP peer.

Table 4: Integrated Haldir process ownership.

Process	Exclusive ownership	Excluded authority
Intent receiver	Intent transport, replay, attachment fetch, and admission records	No policy store or NCP commander credential
Assessment receiver	Assessment ingress, raw evidence, replay, and dispositions	No intent or NCP commander credential
Policy-state authority	Profiles, base policy, intent replay, freshness grants, and policy receipts	No extension transport or NCP commander credential
Commander	Intent conversion, body-issued authority, streams, publication, and reconciliation	No policy evaluation, raw evidence, replay, or profile store

The receivers send only immutable admission records and currentness receipts through narrow authenticated APIs. The commander receives only bounded publication results. Standalone Gate remains a separate deployment mode. It is not a fifth integrated process.

13.3 Fleet, time, and presentation boundaries

The selected X02 profile requires one composite fleet session for each required 1, 2, or 3-drone run. A reusable layout profile fixes generic encoding, bounds, digest, and packing rules. A content-addressed layout instance binds the stable-drone-ID roster, scalar slots, and resources. One aggregate `SensorFrame` contains `6N` ENU position and velocity scalars. One aggregate `CommandFrame` contains `3N` ENU acceleration-command scalars.

The layout assigns one availability group to each drone. N=1 exhausts `00` through `01`. N=2 exhausts `00` through `03`. N=3 exhausts `00` through `07`. Unused high bits are zero. Placeholder storage uses binary64 positive zero, but the decoder exposes only typed unavailability. Zero-based group g maps to command slots $[3g, 3g+1, 3g+2]$ with exact layout-bound restrictive bytes.

The exact source pin retains the bitmap. Each unavailable drone requires its installed three-value restrictive lane in the atomic `3N` Active command. A mismatch rejects the whole command. A correct restrictive Active command receives `APPLIED`, not `HOLD_EFFECTIVE`.

The instance binds each scalar index to its entity, semantic, axis, coordinate frame, unit, encoding, numeric domain, and applicable physical resource. One opaque publisher owns the

layout and its transport slot for each frame class. It assigns the final position, serializes once, and moves the immutable bytes into that slot. The API exposes no detached buffer-to-context handoff. The compact frame adds no slot tag, layout digest, preparation digest, or application authenticator.

TLS record protection detects mutation after seal and before receiver open. It does not attest application provenance. It cannot detect sender changes before seal or receiver changes after open. It cannot infer a plausible same-unit misassociation. Equal values are byte-identical. Bad wiring and a compromised authorized packer also remain outside this claim. Stronger origin claims require independent per-entity attestors.

One NEST 3.9 kernel advances the complete fleet at `0.1 ms` resolution. Each controller epoch is `20 ms`, and the qualification seed is `20260826`. A receipt-bound manifest fixes the NEST build, graph, models, parameters, devices, input transform, decoder, clamps, targets, lane states, threads, seeds, and counter-based streams. Per-drone nodes, connections, devices, and RNG streams are disjoint.

Each lane uses `NORMAL`, `UNAVAILABLE_RESTRICTIVE`, or `RECOVERY_WASHOUT`. Unavailable and washout epochs neutralize only that lane and emit its restrictive output. The first available frame after `UNAVAILABLE_RESTRICTIVE` enters `RECOVERY_WASHOUT`. A second consecutive available frame enters `NORMAL`. Any washout fault returns to `UNAVAILABLE_RESTRICTIVE` and restarts this two-frame recovery. Fault recovery never resets the kernel. Restart restores exact lane state or retires the session. Other lanes remain bitwise equal to their receipt-qualified no-fault baseline.

X02 remains **OPEN**. Historical Host API 2 fleet evidence cannot satisfy native NCP role, transport, closed-loop, or release qualification.

MUSIC alone owns shared-clock simulator coupling. NCP does not implement, tunnel, reinterpret, or replace MUSIC grants, lookahead, barriers, tick ownership, or deadlock handling. The selected X02 profile uses independent clocks and exact NCP source correlation. A shared-clock claim needs a separate MUSIC-qualified deployment and evidence set.

Engram can host verified presentation assets through one bounded local UI ingress. Only its role adapter crosses projects, and it uses NCP. A projection binds an opaque source-receipt digest and grants no protocol authority. SVG is presentation-only and non-contract. SVG cannot carry protocol semantics or runtime evidence.

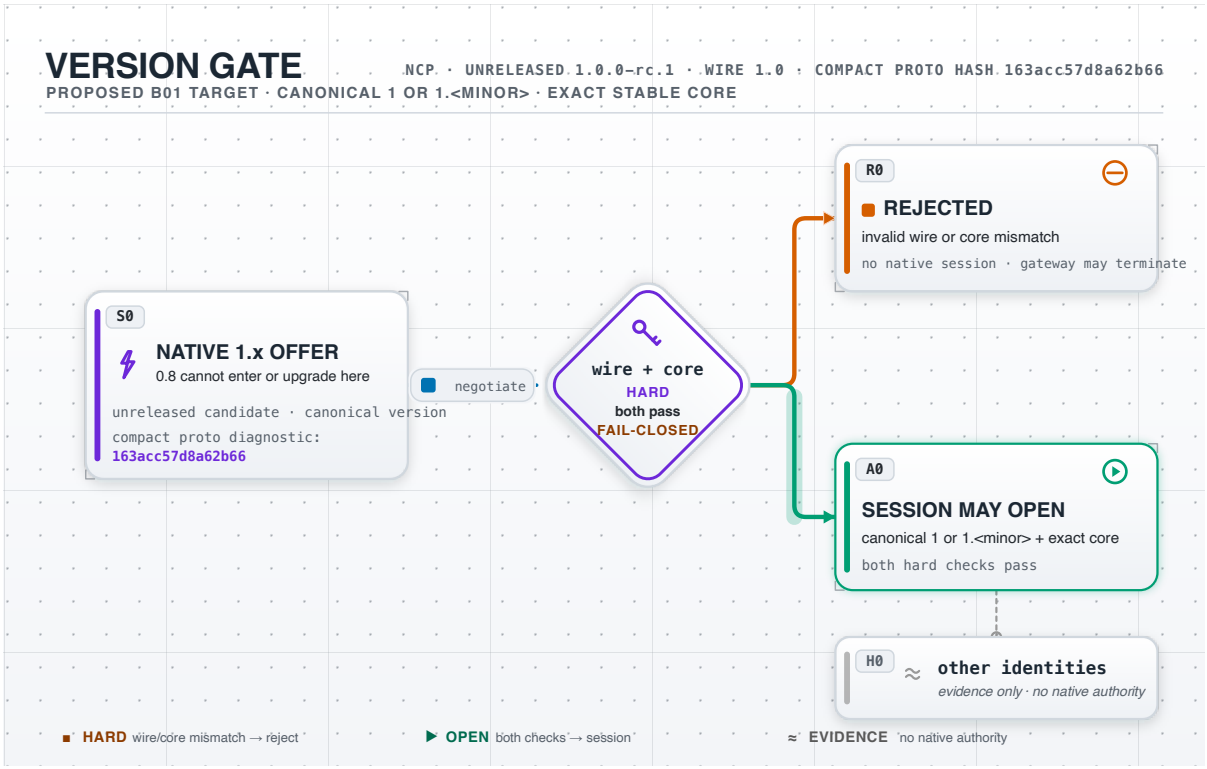

Figure 8: Candidate, stable-core, extension, and release identity domains.

The versioning figure separates wire identity from package, release, extension, and publication identity. The compact protobuf hash is not the complete normative digest. A copied protocol file does not prove migration or installed qualification.

14 Current implementation boundary

The current candidate does not implement the complete target. The most important gaps are:

- Direct `production-secure` Zenoh ingress cannot bind the verified remote principal from the callback surface.
- Callers can construct the current actor helper from supplied evidence. The helper is not an opaque receiver-minted currentness capability.
- The runtime has no one body owner that orders security cuts, leases, command streams, restrictive effects, dispositions, and executor admission.
- The TypeScript client and gateway can remove local generation state before a backend open or close has a definitive result.
- `NeuroControlLoop` stores a host-supplied serialized lease without a receiver-monotonic deadline capability.
- The tick API returns the same command type for local unadmitted fallback and an admitted transport command.
- A send outcome binds a transport position but not the exact payload digest and owner incarnation.
- Stable command wire still defaults a missing mode to HOLD and carries fields that the compact restrictive profiles forbid.

- No accepted compact data-plane encoding, complete no-reuse owner, durable generation issuer, or complete disposition journal exists.
- Compact sensor frames lack mandatory source-bound availability. Zero values cannot represent unavailable sensor groups.
- The TypeScript WebSocket client bounds pending request count but not aggregate bytes.

These are implementation gaps if B01 accepts the target. They are not alternate accepted protocol choices. B02 and B03 remain dependencies for wire-visible implementation.

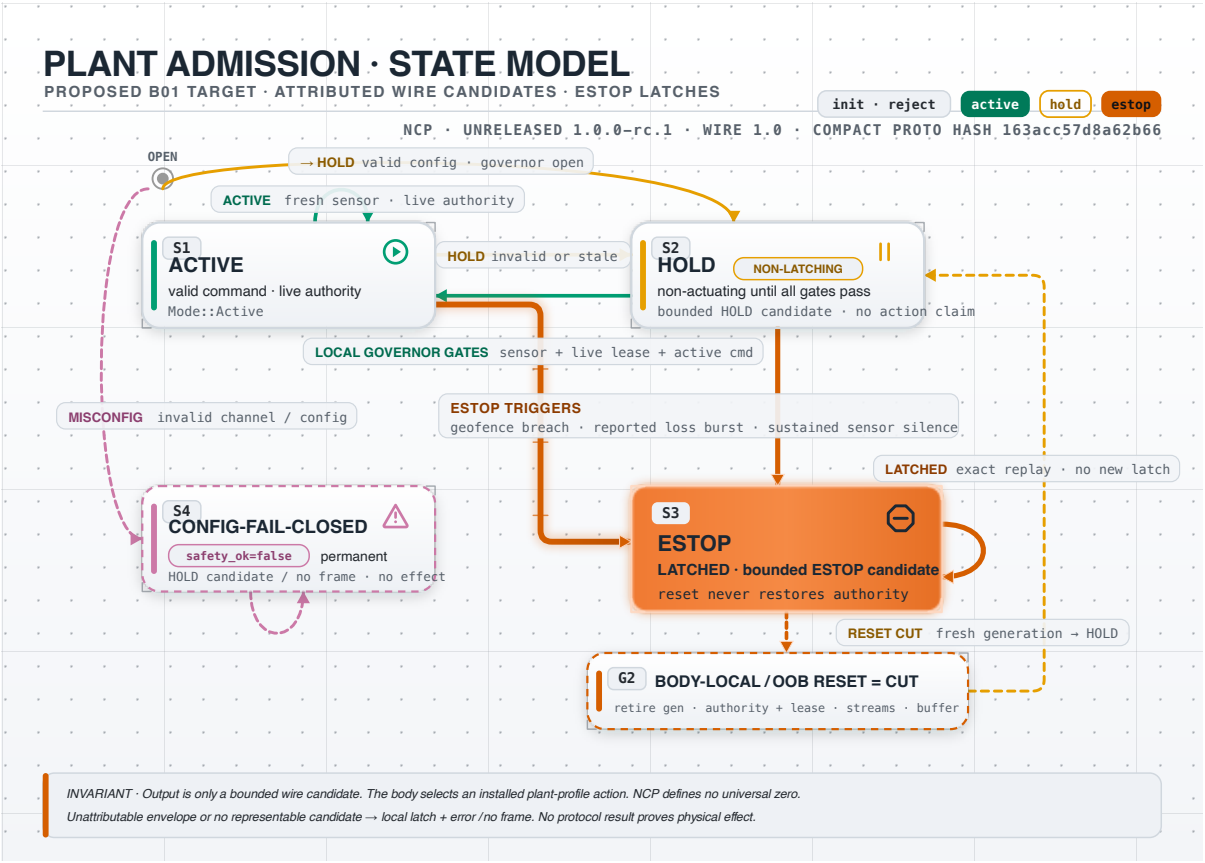

Figure 9: Current compatibility plant-admission state model.

The state model shows the candidate's normalized input, configured actions, restrictive states, and generation-cut reset. It is included to make the current compatibility boundary visible. It is not the complete proposed lifecycle, authority, stream, disposition, and executor owner. The proposed admission ordering remains the target shown earlier in this report.

15 Thirty-lens design review

This review is a maintainer-side critique. It does not satisfy the independent B01 review floor.

Table 5: Thirty-lens review of the proposed target.

Lens	Selected target	Gap or required evidence
Role and type separation	Simulation, plant, observer, extension, and local-library roles are disjoint.	B03 must allocate exact identities. Current wire still uses older shapes.
Identity and correlation	Realm, generation, stream, request, command, and package identities have explicit scopes.	Durable issuers and exact digest domains remain implementation work.
Authentication and authorization	Direct ingress and forwarded signing are default-deny and non-downgrading.	Direct verified-principal binding is unavailable. Live security campaigns are **NOT RUN**.
Error precedence	Raw bounds and profile selection precede identity-sensitive diagnostics.	B03 must freeze exact result codes and precedence.
Freshness and replay	Body-issued absolute grants and per-stream positions fail closed.	Exact grant encoding and digest-bound replay remain open.
Lifecycle and concurrency	One owner orders lifecycle state. Ambiguity retains the same operation.	Cross-process fencing and durable atomicity remain unimplemented.
Resources and denial of service	Every variable count composes with bytes and local allocation charges.	Numeric profiles and load evidence remain open.
Hot-path overhead	Prepared contexts remove repeated parsing. Compact frames decode once.	Compact framing and performance qualification remain **NOT RUN**.
Mathematical consistency	Every equation defines symbols, units, assumptions, and failure behavior.	B03 must select every finite ceiling. Implementations must match boundary tests.
Failure and recovery	Restart grants no Active authority. Unknown effects remain fenced.	Generation, no-reuse, and disposition recovery are not implemented.
Plant safety	The body owns final software admission. Physical effect is outside NCP proof.	Each deployment still needs a plant-owned safety case and interlock.
QoS and overload	Plane capacities are disjoint and finite. Restrictive work has reserved service.	Scheduler values, transport mapping, and installed load tests remain open.
SDK buffering	Pending count and aggregate bytes are both bounded.	Current TypeScript WebSocket buffering has only a count bound.
Extension isolation	Enrolled-store fetch reserves every resource before callback.	B03 must allocate activation, attachment, and parser profiles.
Evolution and compatibility	Stable-core, release, extension, and publication identities remain separate.	B02 must authorize any rebaseline.
Wire and schema parity	Each accepted object must have one source and generated forms.	Current prototype material is not normative parity.

Lens	Selected target	Gap or required evidence
Ecosystem direction	Consumers pin NCP and expose thin adapters.	No installed consumer role is qualified.
Operability	Gaps, drops, expiry, ambiguity, and insecure mode are visible.	Exact metrics and deployment procedures remain open.
Visual traceability	Figures use ordered flow, text labels, redundant cues, and prose descriptions.	Visual review cannot accept an ADR or qualify performance.
Claims and release	Every local result stays inside a named evidence boundary.	Independent review and all external release gates remain unsatisfied.
Sensor missingness	One mandatory bitmap distinguishes unavailable groups from valid zeros. Stable 1.0 fixes polarity, bit order, and padding.	B03 allocates field identity, limits, errors, profiles, and profile-specific placeholders.
Source-conditioned safety	The source pin retains availability and selects profile-specific restrictive lanes.	Body admission and disposition bindings remain unimplemented.
Fleet topology	X02 uses one composite session and atomic command for one to three drones.	Exact rosters and layouts remain unallocated.
NEST execution	One NEST 3.9 kernel advances normal, fault, washout, and recovery epochs.	Real one-, two-, and three-drone evidence is **NOT RUN**.
Authentication profiles	Trusted configuration selects A-direct or B-over-A before bytes.	Direct capability and forwarded JWS implementations remain open.
Language boundary	Opaque capabilities cross Python FFI. Raw credentials and detached buffers do not.	Prepared PyO3 transport remains open.
Standalone and UI boundary	Default Crebain artifacts omit NCP adapters. Presentation hosting carries no runtime semantics.	Consumer packaging and UI tests remain open.
MUSIC separation	NCP owns boundary semantics. MUSIC alone owns shared-clock coupling.	X02 qualifies independent clocks only.
Supply chain and peers	Exact pins, clean archives, SBOMs, and native non-Rust peers bind the final contract.	Publication and clean-room gates remain open.
Operational qualification	Security, faults, rotation, restart, overload, soak, and performance bind exact artifacts.	No ecosystem role campaign has passed.

The review exposes three high-level conclusions. First, low overhead depends on prepared context and strict ownership, not weaker checks. Second, every retry and recovery path needs exact identity. Third, finite counts need byte and allocator bounds.

16 Evidence and verification plan

The publication source and PDF have a narrow verification contract:

1. Convert each generated light SVG to a vector PDF in a disposable build directory.
2. Treat each SVG and converted figure as presentation-only, non-contract material.
3. Build the LaTeX source with fixed time and timezone inputs.
4. Reject LaTeX warnings, overfull boxes, underfull boxes, and undefined references.
5. Require embedded fonts and stable A4 page geometry.
6. Embed and verify a domain-separated SHA-256 commitment over the report, style, and nine SVG sources.
7. Compare the rebuilt PDF with the committed PDF on the same toolchain.
8. Render every page to PNG and inspect layout, labels, equations, tables, and figures.

This contract checks artifact integrity and presentation. It does not validate the protocol or the scientific and safety claims that NCP excludes.

The complete repository gate remains broader. It covers Rust, Python, C, C++, TypeScript, protobuf, schema, profiles, package archives, dependency policy, and candidate baselines. A local pass still cannot satisfy external gates.

The following gates remain separate and **NOT RUN** until exact evidence exists:

- live mTLS, ACL, certificate rotation, and revocation
- two installed and independent non-Rust peers
- fault, soak, duration fuzz, and sanitizer campaigns
- performance qualification
- signatures, SBOM, and provenance
- clean-room reproduction
- all eleven exact consumer and extension role qualifications

17 Equation audit and symbol glossary

Table 6 accounts for every numbered equation in this report. The right-hand column states the interpretation and assumptions. It also states the required failure behavior for unavailable or out-of-domain inputs.

Table 6: Equation audit.

Equations	Unit	Meaning, assumptions, and failure behavior
(1)–(4)	Boolean	The receiver evaluates one fixed gate order. It stops at the first false gate and does not inspect later identity-sensitive fields.

Equations	Unit	Meaning, assumptions, and failure behavior
(5)–(7)	monotonic ticks	Each local stage has ordered receiver ticks. Negative or cross-incarnation durations are invalid. Contiguity is an accounting boundary.
(8)–(10)	monotonic ticks	Adjacent boundary terms cancel in the telescoping sum. Transport, queue, application, and device time are outside this interval.
(11)–(12)	monotonic ticks	Every selected stage ceiling is finite. B03 must select the ceilings, and platform qualification must test them.
(13)–(15)	operations	Per-stage allocation and full-copy counts add. Zero-copy and zero-allocation are fixed-layout targets, not current measurements.
(16)–(20)	bytes	Finite capacities multiply complete per-unit charges. Fixed overhead remains when every occupied count is zero.
(21)	bytes	Products and partial sums use checked arithmetic in a fixed order. Overflow rejects the profile before allocation.
(22)	local charge	The implementation applies a second finite monotone accounting domain. JavaScript reports an abstract charge unless it controls the complete representation.
(23)–(32)	elements and bytes	Current bounds and deterministic allowable victim selection define the survivor state. Checked reservation establishes the post-admission invariants. Disjoint lifecycle and queue charges then form one portable prepared-memory bound.
(33)–(35)	local charge	Each queue has a finite monotone local charge. Disjoint lifecycle and queue charges add without cross-plane borrowing.
(36)–(38)	operations	One exact namespace has at most one active lifecycle record. A cross-process owner also needs a fenced writer epoch.
(39)–(41)	ordered state transitions	Reservation precedes external work. Result publication and high-water commit precede active-slot release.
(42)–(44)	bytes and Boolean	Retry requires exact coordinate, request, context, and ordinal equality. Any unequal component is a conflict.
(45)–(48)	ordinal	Zero is the initial high-water sentinel and is never assigned. Checked succession is strictly monotone. Exhaustion seals the epoch instead of wrapping or saturating.
(49)	service count	Session state uses bounded shared owners. The design allocates no thread, socket, or store service per session.
(50)–(55)	position	Position zero is the initial high-water sentinel and is never assigned. Checked succession is strictly monotone. Exhaustion seals the epoch. Assignment commits before later work, so failure creates a visible gap and never permits reuse.

Equations	Unit	Meaning, assumptions, and failure behavior
(56)–(60)	bytes and work units	A prepared fixed layout bounds group count, bitmap bytes, value count, and exact frame length. Linear work is a structural count, not elapsed time.
(61)–(67)	requests, bytes, and local charge	Admission leaves one free request slot. Every coefficient is a deployment maximum. Each window reserves portable and local storage before backend work. The active-session capacity lifts both bounds across the deployment.
(68)–(70)	position, digest, and bytes	Only the next exact position executes. Completion advances by checked succession. Response release also requires exact retained request bytes.
(71)–(74)	ticks and Boolean	One receiver clock and body-issued grant define an exclusive deadline. Every currentness, lease, grant, stream, profile, and executor predicate must be true.
(75)	proof relation	NCP does not prove that software admission implies physical effect. The equation does not assert that an effect is impossible.
(76)–(77)	bytes and items	One complete envelope and its external attachment references remain within installed count and byte ceilings.
(78)–(82)	bytes and local charge	One activation reserves the complete envelope plus the larger active or terminal metadata charge. Summation over the bounded activation count lifts both accounting domains across the deployment.
(83)–(86)	bytes and local charge	Auxiliary arenas have independent deployment ceilings. Adding every disjoint owner produces one complete NCP-owned bounded-state envelope.
(87)	graph edge	Consumers depend on NCP through thin adapters. NCP does not import consumer application code.

Table 7: State and result terms.

Term	Exact meaning
Current	The installed receiver-owned state still admits the checked operation
Pending	Capacity and lineage are durable, and external work can be active
Ambiguous	The same operation remains fenced because effect outcome is unknown
Terminal	A definitive result and high-water are durable
Retired	Reexecution is forbidden, while the full retained result can be unavailable
Exact retry	Coordinate, request bytes, context bytes, and ordinal all equal retained values
Software admission	The body executor slot accepted the exact command and evidence transition
Physical effect	A deployment result outside the NCP software proof boundary

18 Conclusion

The proposed architecture obtains low overhead from preparation, exact ownership, one-pass decoding, finite shared stores, and plane isolation. It does not remove authentication, authorization, freshness, safety, or currentness checks.

The equations expose the design accounting used here. Latency is a telescoping sum of named local stages. Memory is the sum of fixed overhead and capacity-scaled charges. Lifecycle mutation is single-flight. Retry is exact equality. Ordinals and positions use checked monotone successors. Queue and extension work reserve complete byte and metadata budgets before retention.

The target remains proposed. Independent B01 review, B02 authorization, B03 allocation, complete implementation, conformance, installed peers, and every external gate still remain.